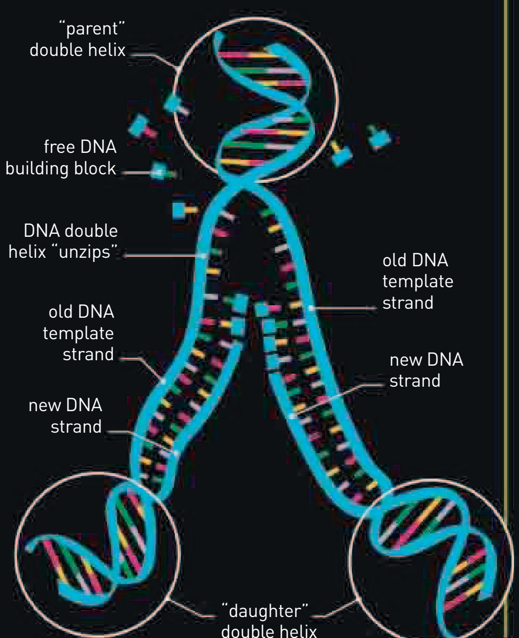
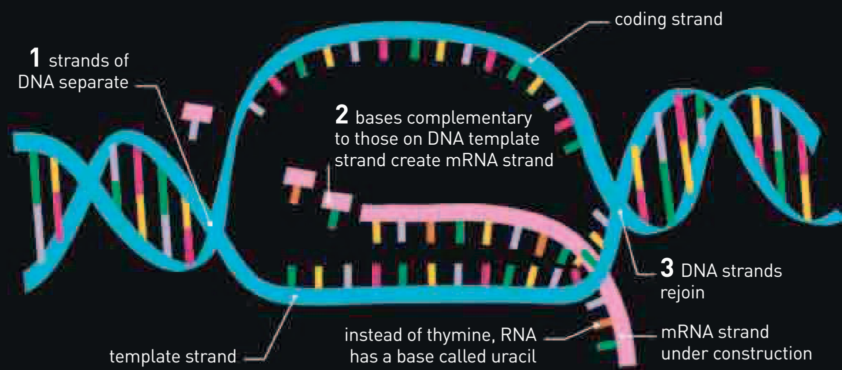
DNA REPLICATION

Just before a cell divides, it replicates its entire DNA. Each DNA molecule "unzips," and the paired bases separate. Because of the strict base-pair ruling, the base sequence along one strand determines the sequence along the other: they are complementary. Free DNA building blocks are linked to make new complementary strands along each existing strand "template." This creates material for two new, genetically identical double helices. At cell division, one double helix goes to one cell, and one goes to the other.

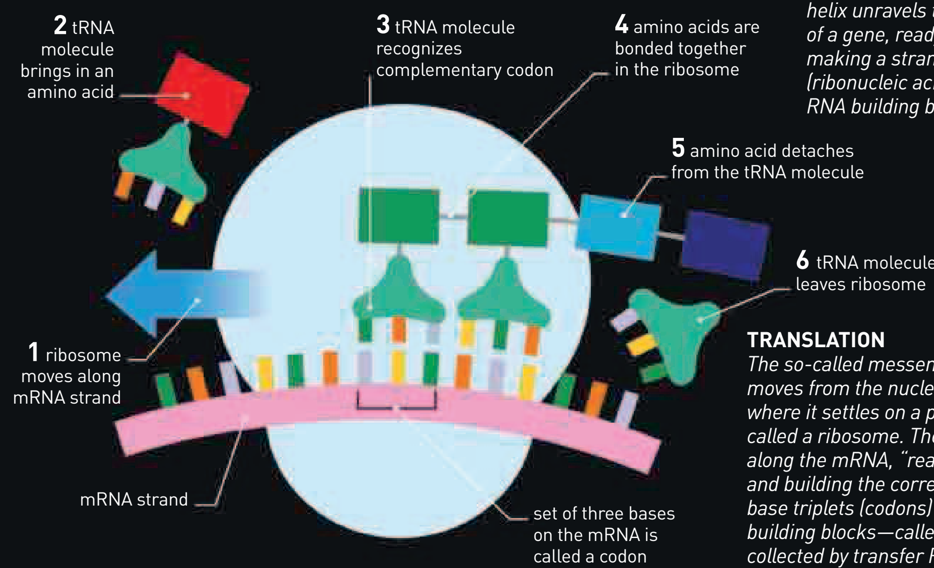


MAKING PROTEINS

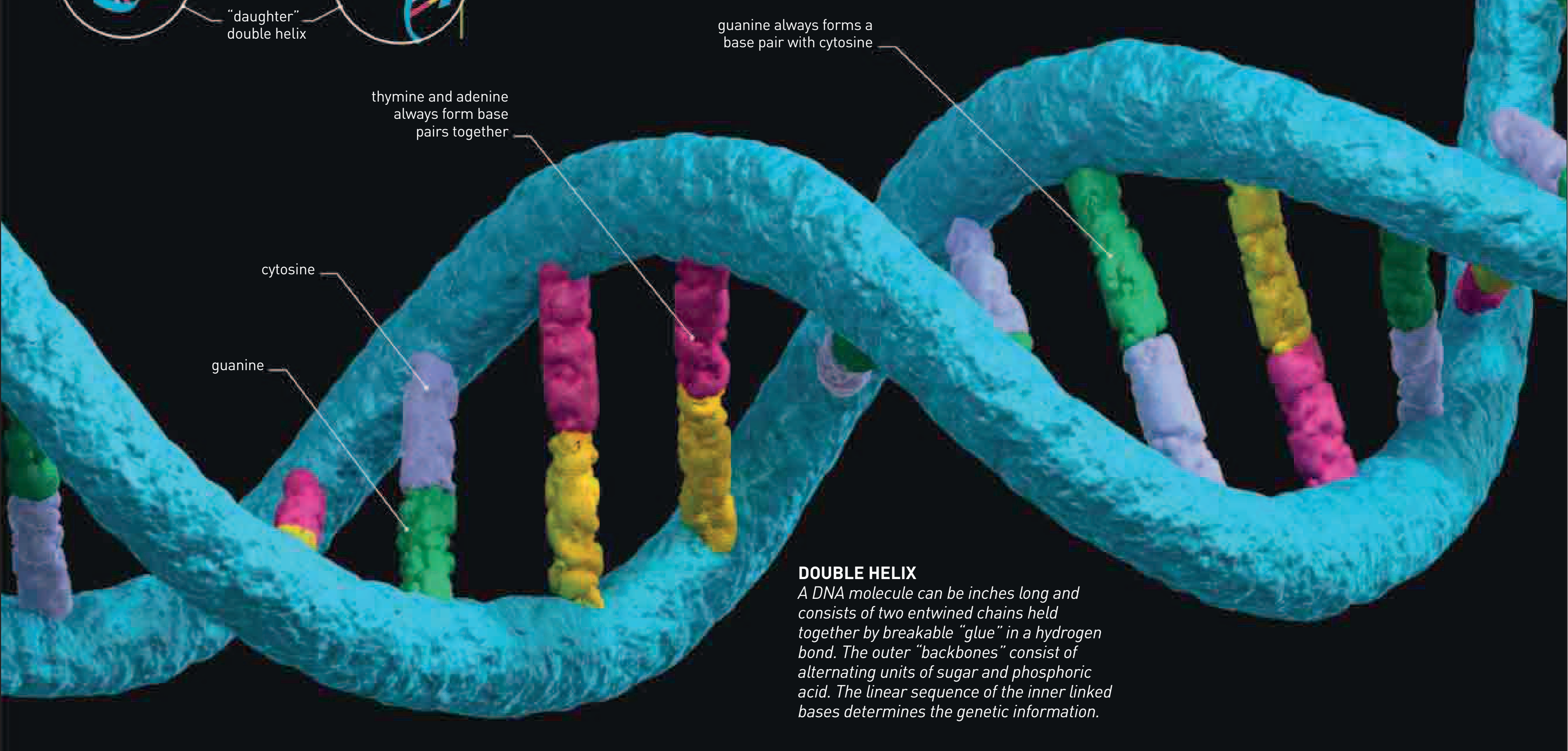
A gene is a section of DNA containing instructions to assemble a protein molecule to carry out a specific task, such as making a pigment. In this way, genes determine an organism's characteristics. Before a protein is assembled, the gene's base sequence must be "copied" in the cell nucleus, a process called transcription. This copy is then sent to the cytoplasm. In another process, called translation, the base sequence information is used to assemble the protein.



TRANSCRIPTION
Inside the nucleus, part of a DNA double helix unravels to expose the coding region of a gene, ready for "copying." This involves making a strand of nucleic acid called RNA (ribonucleic acid) by bonding together free RNA building blocks.



TRANSLATION
The so-called messenger RNA (mRNA) moves from the nucleus to the cytoplasm, where it settles on a protein-making granule called a ribosome. The ribosome moves along the mRNA, "reading" its base sequence and building the correct protein. Specific base triplets (codons) dictate specific protein building blocks—called amino acids—collected by transfer RNA (tRNA) molecules.



DOUBLE HELIX
A DNA molecule can be inches long and consists of two entwined chains held together by breakable "glue" in a hydrogen bond. The outer "backbones" consist of alternating units of sugar and phosphoric acid. The linear sequence of the inner linked bases determines the genetic information.